

APPEND ALL THREE DATASET AND CREATE ONE RWA DATASET

1 — Import Library

This cell imports the pandas library, which is used for reading, handling, and processing datasets in Python.

```
In [1]: import pandas as pd
```

2 — Load All Three Datasets

This cell loads the Adani, Reliance, and TCS stock datasets from CSV files into pandas DataFrames for further processing.

```
In [3]: # Load three datasets

adani = pd.read_csv("adani_stock.csv")
reliance = pd.read_csv("reliance_stock.csv")
tcs = pd.read_csv("tcs_stock.csv")
```

3 — Add Company Column

This cell adds a new column named Company to each dataset so that after combining the datasets, we can identify which row belongs to which company.

```
In [4]: # add company column

adani['Company'] = 'Adani'
reliance['Company'] = 'Reliance'
tcs['Company'] = 'TCS'
```

4 — Append All Three Datasets

This cell combines all three datasets into one single dataset using `pd.concat()`. The datasets are appended row-wise because all datasets have the same structure and columns.

```
In [5]: # append the datasets

combined_stock_data = pd.concat(
    [adani, reliance, tcs],
    axis=0,
    ignore_index=True
)
```

5 — Check Combined Dataset

This cell displays the first 5 rows of the combined dataset to verify that the append operation worked correctly.

```
In [6]: # check combine dataset

combined_stock_data.head()
```

```
Out[6]:
```

	Price	Close	High	Low	Open
0	Ticker	ADANIENT.NS	ADANIENT.NS	ADANIENT.NS	ADANIENT.NS
1	Date	NaN	NaN	NaN	NaN
2	2015-01-01	70.76136779785156	71.12961809193881	69.40390937432629	70.0970796331477
3	2015-01-02	71.10795593261719	71.77224405824526	70.76136363541752	70.94910442431483
4	2015-01-05	72.28491973876953	73.15138269048227	70.86247279895221	70.92023699573306

6 — Check Dataset Shape

This cell shows the total number of rows and columns in the combined dataset after appending all three datasets.

```
In [7]: #check shape

print(combined_stock_data.shape)
```

```
(7407, 7)
```

7 — Save Appended Dataset

This cell saves the final combined raw dataset into a new CSV file named combined_stock_dataset_raw.csv for further cleaning, visualization, and machine learning tasks later.

```
In [8]: # save dataset

combined_stock_data.to_csv(
    "combined_stock_dataset_raw.csv",
    index=False
)
print("Raw combined dataset saved successfully!")
```

```
Raw combined dataset saved successfully!
```

```
In [ ]:
```